

Land Allocation Modeling with New Theory Principles (MMC051)

July 19, 2026

1 Introduction

The "New Theory" is a concept introduced by King Rama IX of Thailand emphasizing self-sustenance for farmers. The prominent recommendation for dividing standard 1-rai (1600 m^2) land is

- 30% water resources zone
- 30% rice farming zone
- 30% mixed crops cultivation zone
- 10% residential & utility zone

The standard 1-rai land refers to the land in which is flat, square, and with equally distributed sunlight, water, and access. In actuality, land is frequently imperfect and irregular. Furthermore, elevation and water source location both contribute to land division. Therefore, the "New theory" should be treated as a guiding principle, and not a factual blueprint.

In this problem, we are given a 1-rai quadrilateral land, and our tasks are to

1. Calculate the land and dimensions geometrically
2. Assess the land across various factors including water availability, sunlight, and access to roadway.
3. Establish mathematical model(s) to properly allocate the land across categories of usage (zone).

2 Key Consideration and Assumptions

The following assumptions are to be made, as per the problem statement:

1. The area of the land is 1-rai or $1,600\text{ m}^2$
2. The shape of the land is an asymmetrical quadrilateral
3. The elevation of the land slopes downward, from westernmost highland to lowland in the east.

4. The pond must store an adequate amount of water for year-around agricultural use.
5. The residential & utility zone must be located on the highland and closely connected to the roadway.

Moreover, in these model(s), we herein use these additional assumptions

1. Assume that the incline of elevation is constant, where the angle of inclination is denoted as γ
2. Assume that a single roadway is located adjacent to one side of the land in parallel with its border.
3. Assume that the quadrilateral is convex, i.e., each of its interior angle is less than 180°
4. Assume that the rice is cultivated once a year during a 120-day duration, from July to October, when the rainy season lies.

3 Model Formation

According to the New Theory, let the land be divided into 4 zones: water resources (zone 1), rice farming (zone 2), mixed crops cultivation (zone 3), and residential & utility (zone 4)

Furthermore, we define the following variables as

1. A_i is the area (m^2) of the i^{th} zone.
2. W_i is the weighting coefficient, indicating the importance of the i^{th} zone.
3. E_i is the efficiency of the i^{th} zone, scaled from 0 to 1, inclusive.

Then, we obtain the core model for the total efficiency of the division as shown:

$$\text{Total Efficiency} := \sum_{i=1}^4 A_i W_i E_i$$

Our objective is to maximize the value of total efficiency.

3.1 Parameters

Our parameters are the geographical characteristics of the land. (x_a, y_a) , (x_b, y_b) , (x_c, y_c) , and (x_d, y_d) respectively represent the cartesian coordinates of the quadrilateral vertices, denoted by A, B, C , and D . Without loss of generality, we will fix A at the origin and $\overline{AB}$ to be on the x -axis.

The compass direction is represented as the angle $\theta \in [0^\circ, 360^\circ)$ between the y -axis and the north direction (anti-clockwise). Recall that γ is the angle of steepness of the land and it is given in the problem statement that the land slopes downwards from west to east.

Next, The road direction is denoted as D_R , and the direction of external water source is denoted as D_W . D_R being equal to 1, 2, 3, and 4 indicates the edge adjacent to the road, $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, and $\overline{DA}$, respectively. D_W holds similar properties.

Finally, we define R_{in} as the rate of water intake. We assume that this rate is a constant. So, our parameter is a nonuple

$$(A, B, C, D, \theta, \gamma, D_R, D_W, R_{in})$$

3.2 Decision Variables

We overlay the land into grids of size $1 \text{ m} \times 1 \text{ m}$, intersected within the land's boundaries. The i^{th} grid is represented by a quadruple N_i , defined by

$$N_i := (x_i, y_i, T_i, s_i)$$

where

1. (x_i, y_i) is the cartesian coordinate of the grid's centroid. (The cartesian unit is meter throughout this report.)
2. T_i indicates the type of zone which corresponds to the grid; i.e. $T_i = k$ indicates that the grid N_i belongs to k^{th} zone.
3. s_i is the area of the grid (square meter).

So, our decision variables is

$$(N_1, N_2, N_3, N_4, \dots).$$

Note: Some grids, such as the edges, might not have the area of $1 \text{ m} \times 1 \text{ m}$ nor the shape of a quadrilateral.

3.3 Constants

In this report, there are many constants, for instances, weights (importance), rates of water use per area, and energy from the Sun. Recall that these

constants are not parameters. In the model formation section, they will be left as variables, which will be defined in each corresponding sections, and determined in section 4.

3.4 Section Area, Center of Mass, Height, and Average Height

Let h_i be the relative height at the centroid position of the i^{th} grid compared to the lowest easternmost elevation, hence, h_i can be formulated using a 2D-rotation of the points into the north-east coordinates and modelled using the assumed linear steepness as follows,

$$h_i := x_i \sin \theta \tan \gamma + y_i \cos \theta \tan \gamma$$

Note: We will be referring to this relative height as simply the "height".

For $i = 1, 2, 3, 4$, we define the functions $\alpha_i : \mathbb{N} \rightarrow [0, 1]$ and $\beta_i : \mathbb{N} \rightarrow \{0, 1\}$ such that

$$\alpha_i(k) := \begin{cases} s_k & \text{if } T_k = i \\ 0 & \text{otherwise} \end{cases}$$

$$\beta_i(k) := \begin{cases} 1 & \text{if } T_k = i \\ 0 & \text{otherwise} \end{cases}.$$

Recall that A_i is the area of zone i . Let $x_{\text{CM},i}$, $y_{\text{CM},i}$, and $\bar{h}_i$ be the coordinate of the center of masses in the x and y axis and the average height of zone i (meter), respectively. Therefore,

$$A_i := \sum_k \alpha_i(k)$$

$$x_{\text{CM},i} := \frac{\sum_k x_i \alpha_i(k)}{A_i}$$

$$y_{\text{CM},i} := \frac{\sum_k y_i \alpha_i(k)}{A_i}$$

$$\bar{h}_i := \frac{1}{\sum_k \beta_i(k)} \sum_k h_k \beta_i(k)$$

The center of mass is chosen to represent each section as it acts as the weighted average balance point of that respective section. Thus, some variables will be expressed in terms of the values at the centroid position, i.e. section distances.

3.5 Water Resources (Zone 1)

3.5.1 Water Mechanics

Recall that R_{in} is the rate of water intake (cubic meters per day). We define R_{out} as the rate of water usage (cubic meters per day). For sustainability, R_{in} must not be lower than R_{out} , otherwise the farm will eventually run out of water.

We assume that R_{out} varies proportionally to the area of rice farming and mixed crops cultivation zone. Define U_2 and U_3 respectively as the rates of water use per area in zone 2 and 3, which are constant. Recall that A_2 and A_3 are the area of zone 2 and zone 3, respectively. Hence,

$$R_{\text{out}} := U_2 A_2 + U_3 A_3.$$

Thus, we have the constraint

$$R_{\text{in}} \geq U_2 A_2 + U_3 A_3$$

3.5.2 Volume

The possible volumes of the water resource also depends on the surface area. According to Clermont County Soil and Water, the water resource should have side slopes of 3:1 ratio (three feet out, one foot down). This formula is then used to calculate the volume V (cubic meter) of the water resource.

$$V := \sum_{N_i} v(N_i)$$

where

$$v(N_i) := \begin{cases} \frac{1}{3}; & \text{if the grid } N_i \text{ is an edge or a corner grid} \\ & \text{of water zone} \\ \min(\mathcal{N}_i) + \frac{1}{3}; & \text{otherwise} \end{cases}$$

while $\mathcal{N}_i := \{v(N_j) | N_j \text{ is a grid surrounding } N_i \text{ in any way including vertically, horizontally, and diagonally}\}$.

For sustainability, we suggest that the volume of the water resource must provide enough water for 60 days in case of emergency. So, we have the constraint

$$V \geq 60R_{\text{out}} = 60U_2 A_2 + 60U_3 A_3$$

It would be more sustainable if the water resource can provide enough water for an even longer period. However, too large water resource is unnecessary and space-inefficient. Let day_{opt} be the optimal duration for the water resource to provide water for. Therefore, the efficiency regarding the sufficiency of water, denoted as $E_{w,1}$, is defined as

$$E_{w,1} := e^{-k_{w,1}(\text{day}_{\text{actual}} - \text{day}_{\text{opt}})^2}$$

when $\text{day}_{\text{actual}} = \frac{V}{U_2 A_2 + U_3 A_3}$ and $k_{w,1}$ is a constant.

3.5.3 Connection to External Water Source

Recall that D_W indicates the direction of the external water source. The farm's water resource must have at least one grid that borders the land boundary in the external water source direction. This is a constraint.

3.5.4 Efficiency

The efficiency of the water resource E_1 is determined by $E_{w,1}$, so

$$E_1 := e^{-k_{w,1}(\text{day}_{\text{actual}} - \text{day}_{\text{opt}})^2}$$

3.6 Rice Farming (Zone 2)

3.6.1 Distance to Water Resources

Let $d_{2,1}$ be the cartesian distance in meters between the center of masses of the rice field and the water resource. Hence,

$$d_{2,1} = \sqrt{(x_{\text{CM},2} - x_{\text{CM},1})^2 + (y_{\text{CM},2} - y_{\text{CM},1})^2}$$

By design, paddy rice fields should be near a water source and at a relatively low elevation due to flooded fields and water requirements. Hence, we prioritize on **mimimizing** the distance of the center of masses between the rice field and the water resource. Therefore, the efficiency regarding distance (linearly), denoted as $E_{d,2}$, is defined as

$$E_{d,2} := 1 - \left| \frac{d_{2,1} - d_{\min}}{d_{\max} - d_{\min}} \right|.$$

where $d_{\min} = 0$ and $d_{\max}$ is the longest diagonal of the land,

$$d_{\max} = \max \left(\sqrt{(x_a - x_c)^2 + (y_a - y_c)^2}, \sqrt{(x_b - x_d)^2 + (y_b - y_d)^2} \right)$$

3.6.2 Elevation

Let $E_{h,2}$ be the ratio between the field's average height and the maximum possible height.

For suitability, we must **maximize** this value alongside the field's average elevation, $\bar{h}_2$, to ensure proper field flooding and higher yields. Therefore,

$$E_{h,2} := 1 - \frac{\bar{h}_2}{h_{\max}}$$

3.6.3 Light

Thailand lies in the northern hemisphere, hence the sun's declination is towards the South until the rainy season ends, which is when rice cultivation usually take place. Therefore, on average, the rice grown on the northern side of the residential area tends to be blocked by its shadow for a longer period in the afternoon, thus grow less effectively than rice grown on the southern side.

Let ϕ_2 be the clockwise angle between vertical axis at the residential area center of mass, and the line connecting the center of masses of the rice field and the residential area. Thus, we must **maximize** ϕ_2 in order for rice to grow effectively.

$$\phi_2 := 90^\circ + \arctan\left(\frac{y_{CM,4} - y_{CM,2}}{x_{CM,4} - x_{CM,2}}\right)$$

The efficiency regarding lighting, denoted as $E_{l,2}$, is defined as

$$E_{l,2} := \frac{\phi_2}{\phi_{\max}}$$

where $\phi_{\max} = 180^\circ$ indicates the rice field is directly to the south of the residential area.

3.6.4 Efficiency

The efficiency for rice farming, denoted as E_2 , is defined by

$$E_2 := \frac{W_{d,2}E_{d,2} + W_{h,2}E_{h,2} + W_{l,2}E_{l,2}}{W_{d,2} + W_{h,2} + W_{l,2}}$$

$$E_2 = \frac{W_{d,2}\left(1 - \left|\frac{d_{2,1} - d_{\min}}{d_{\max} - d_{\min}}\right|\right) + W_{h,2}\left(1 - \frac{\bar{h}_2}{h_{\max}}\right) + W_{l,2}\left(\frac{\phi_2}{\phi_{\max}}\right)}{W_{d,2} + W_{h,2} + W_{l,2}}$$

when $W_{d,2}, W_{h,2}, W_{l,2}$ are weights.

3.7 Mixed Crops Cultivation (Zone 3)

3.7.1 Distance to Water Resource

Let $d_{3,1}$ be the cartesian distance in meters between the center of masses of the mixed crop cultivation and the water resource.

$$d_{3,1} = \sqrt{(x_{CM,3} - x_{CM,1})^2 + (y_{CM,3} - y_{CM,1})^2}$$

Similarly, the efficiency regarding distance, denoted as $E_{d,3}$, is defined as

$$E_{d,3} := 1 - \left|\frac{d_{3,1} - d_{\min}}{d_{\max} - d_{\min}}\right|.$$

3.7.2 Elevation

Let $E_{h,3}$ be the ratio between the mixed crop's average elevation and the optimal mixed crop's elevation.

For suitability, we must **maximize** the distance between this value alongside the field's average elevation and the optimal elevation, $\bar{h}_3$, to ensure proper irrigation (not too high or low) and higher yields. Therefore, the efficiency regarding height, denoted as $E_{h,3}$, is defined as

$$E_{h,3} := e^{-k_{h,3}(\bar{h}_3 - h_{\text{opt}})^2}$$

where $h_{\text{opt}} = \frac{h_{\max} - h_{\min}}{2}$, which makes $E_{h,3}$ returns the value of 0.5 to 1, inclusive. The value approaches 1 as $\bar{h}_3$ approaches h_{opt} .

3.7.3 Light

This criteria for mixed crops is the same as rice fields. Though, the importance of light towards growth efficiency for mixed crops should be less than those for rice fields, this will be reflected in the weights. So similar to ϕ_2 , ϕ_3 is as follows,

$$\phi_3 := 90^\circ + \arctan\left(\frac{y_{CM,4} - y_{CM,3}}{x_{CM,4} - x_{CM,3}}\right)$$

The efficiency regarding lighting, denoted as $E_{l,3}$, is defined as

$$E_{l,3} := \frac{\phi_3}{\phi_{\max}}$$

3.7.4 Efficiency

The efficiency for mixed crops cultivation, denoted as E_3 , is defined by

$$E_3 := \frac{W_{d,3}E_{d,3} + W_{h,3}E_{h,3} + W_{l,3}E_{l,3}}{W_{d,3} + W_{h,3} + W_{l,3}}$$

$$E_3 = \frac{W_{d,3}\left(1 - \left|\frac{d_{3,1} - d_{\min}}{d_{\max} - d_{\min}}\right|\right) + W_{h,3}\left(e^{-k_{h,3}(\bar{h}_3 - h_{\text{opt}})^2}\right) + W_{l,3}\left(\frac{\phi_3}{\phi_{\max}}\right)}{W_{d,3} + W_{h,3} + W_{l,3}}$$

when $W_{d,3}, W_{h,3}, W_{l,3}$ are weights.

3.8 Residential & Utility (Zone 4)

3.8.1 Elevation

Let $E_{h,4}$ be the ratio between the resident's average elevation and the maximum possible elevation.

For suitability, we must **maximize** this value alongside the resident's average elevation, $\bar{h}_4$, to ensure safety from flooding and landslides. Therefore,

$$E_{h,4} := \frac{\bar{h}_4}{h_{\max}}$$

3.8.2 Connection to Road

Based on our key assumptions, the residential area must be closely connected to the roadway. Therefore, the residential area must have at least one grid that borders the land boundaries in the road direction, otherwise, that combination is implausible. This is a rucial constraint.

3.8.3 Efficiency

The efficiency of residential & utility E_4 is from the height efficiency $E_{h,4}$, so

$$E_4 := \frac{\bar{h}_4}{h_{\max}}$$

3.9 Additional Constraints

Each zone, excluding the water resource, must have a minimum required area. We set the minimum constraints as

$$A_2 \geq 300, A_3 \geq 200, A_4 \geq 100$$

3.10 Model Summary

We have the model

$$\text{Total Efficiency} = A_1 W_1 E_1 + A_2 W_2 E_2 + A_3 W_3 E_3 + A_4 W_4 E_4$$

with W_i being weights and $A_i = \sum_k \alpha_i(k)$ for $i = 1, 2, 3, 4$ and

$$E_1 = e^{-k_{w,1}(\text{day}_{\text{actual}} - \text{day}_{\text{opt}})^2}$$

$$E_2 = \frac{W_{d,2} \left(1 - \left| \frac{d_{2,1} - d_{\min}}{d_{\max} - d_{\min}} \right| \right) + W_{h,2} \left(1 - \frac{\bar{h}_2}{h_{\max}} \right) + W_{l,2} \left(\frac{\phi_2}{\phi_{\max}} \right)}{W_{d,2} + W_{h,2} + W_{l,2}}$$

$$E_3 = \frac{W_{d,3} \left(1 - \left| \frac{d_{3,1} - d_{\min}}{d_{\max} - d_{\min}} \right| \right) + W_{h,3} \left(e^{-k_{h,3}(\bar{h}_3 - h_{\text{opt}})^2} \right) + W_{l,3} \left(\frac{\phi_3}{\phi_{\max}} \right)}{W_{d,3} + W_{h,3} + W_{l,3}}$$

$$E_4 = \frac{\bar{h}_4}{h_{\max}}$$

Moreover, we also obtained the constraints as following:

1. $R_{in} \geq U_2 A_2 + U_3 A_3$.
2. $V \geq 60 U_2 A_2 + 60 U_3 A_3$.
3. $A_2 \geq 300$.
4. $A_3 \geq 200$.
5. $A_4 \geq 100$.
6. The farm's water resource must have at least one grid that borders the land boundary in the external water source direction.

7. The residential area must have at least one grid that borders the land boundary in the road direction.

The model will return the value of A_1, A_2, A_3, A_4 that maximizes the total efficiency. The proportion of the i -th region could be calculated using formula

$$\% \text{ Proportion of } i\text{-th Region} = \frac{A_i}{A_1 + A_2 + A_3 + A_4} \times 100\%$$

4 Constants Determination

4.1 Analytical Hierarchy Process (AHP)

After we have determined separate 0-1 scale indices, we used the **Analytical Hierarchy Process (AHP)** to obtain suitable weights for our combined indices. i.e. E_2, E_3 , and the Total Efficiency, by conducting pair-wise comparisons of relative importance between each two factors. This results in a decision matrix based on our weightings, given the sustainability and suitability goals.

Therefore, the calculated weights using the AHP and decision matrices (see Appendix A) are as follow,

Weight	Value
$W_{d,2}$	0.3119
$W_{h,2}$	0.4905
$W_{l,2}$	0.1976
$W_{d,3}$	0.5485
$W_{h,3}$	0.2409
$W_{l,3}$	0.2106
W_1	0.3416
W_2	0.3774
W_3	0.1405
W_4	0.1258

4.2 Water Usage Constants

The U_2 and U_3 constants from section 3.5.2 are determined accordingly to the information from the Rice Department and the Department of Agricultural Extension. We obtain

$$U_2 = 0.0090 \text{ and } U_3 = 0.0045.$$

4.3 Optimal Storage Volume Constants

The day_{opt} value from section 3.5.2 is set as 365 days. We are going to determine the value of $k_{w,1}$ from section 3.5.2. We expect that the efficiency should be 0.5 when $\text{day}_{\text{actual}} = 60$, thus

$$e^{-k_{w,1}(60-365)^2} = e^{\ln 0.5}$$

$$k_{w,1} = -\frac{\ln(0.5)}{305^2} \approx 7.4511 \times 10^{-6}$$

4.4 Optimal Height of Crop Cultivation Constant

First, we select h_{opt} in 3.7.2 to be half of h_{max} . We are going to determine the value of $k_{h,3}$ constant from section 3.7.2. We expect that the function should return 0.5 when $\bar{h}$ is furthest from $h_{\text{opt}} = \frac{h_{\text{max}}}{2}$. So,

$$\bar{h} = h_{\text{max}} \text{ or } 0$$

Substitute $k = h_{\text{max}} = 2h_{\text{opt}}$,

$$e^{-k_{h,3}(2h_{\text{opt}}-h_{\text{opt}})^2} = e^{\ln 0.5}$$

$$k_{h,3} = -\frac{\ln 0.5}{h_{\text{opt}}^2}$$

5 Computation and Optimization

The entire computation and optimization process is completed using C++ programming language, refer to Appendix B for the full code snippet.

The 1-rai land is overlaid with grids of size 1 m $\times$ 1 m, each grid centroid is located at lattice point in cartesian coordinate system, in other word, all grid centroid are located on a point where all coordinate values are integers. The grids are generated via Ray-casting algorithm. The objective of our simulation is to maximize the total efficiency given the obtained weights from AHP and the constants from real-world data.

Its input is the vertices of the land plot, D_R , D_W , θ , and γ . The initial composition of the land (zone combinations) is determined at random with a fixed starting ratio $A_1 : A_2 : A_3 : A_4$. Afterwards, our algorithm utilizes the concept of the **hill climbing algorithm combined with movements to center-of-mass prioritization**, where random local search make comparisons between a pair of grids, and determines the optimal composition of scenarios in which these two grids are swapped or not, and decide whether to swap the pair or not. The total efficiency is computed for the land everytime, until a given number of swap iterations is reached, and the program terminates.

Many optimization and algorithmic techniques are implemented alongside the code including but not limited to

- The determination of whether each grid is situated inside the land was done using ray-casting algorithm, employing vector cross product.
- mersenne-twister based pseudo-random number generator, specifically mt19937 was used for randomness generation.

- To determine the volume of water source, we use multi-source Breadth-first Search algorithm.
- Vectorization in C++ was used, specifically GNU g++ compiler pragmas, to auto-vectorize loops, use avx2 instruction sets, and optimize floating-point arithmetic.
- To increase the speed, caching techniques were used for regularly access values, such as trigonometry function values.

The computation are done with different ratio values (sevenfold for each ratio), each ratio differs by 0.1. After the search is done, we then computed for different quadrilateral shapes. We first begin with a perfect 40 m $\times$ 40 m square with $\theta = 0^\circ$ and $\gamma = 10^\circ$ as our baseline assessment, then we later applied asymmetric and irregular quadrilateral shapes, which is applicable to real-world landscapes.

6 Example Results

First, we conducted our model on the generic square grid (40 m $\times$ 40 m) as our baseline assessment, by using $\theta = 0^\circ$, $\gamma = 10^\circ$, and $R_{\text{in}} = 300$, comparing the total efficiency at the most optimal point, the selected results (excluding ratios that defy the constraints) are as follows (refer to Appendix C for the full simulation):

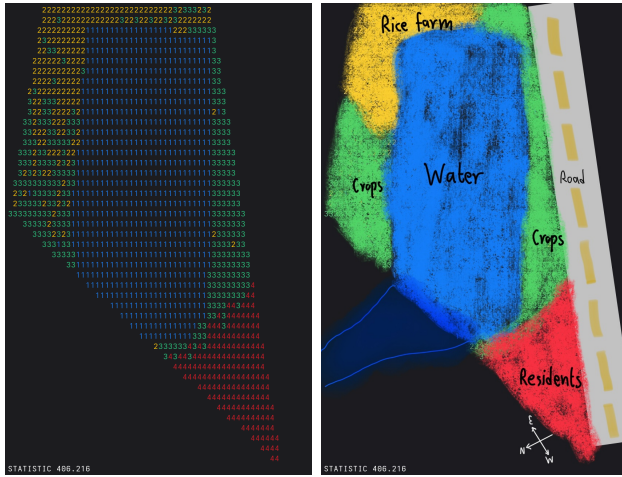
% A_1	% A_2	% A_3	% A_4	Max Total Efficiency
20	20	20	40	260.213
20	20	30	30	258.612
30	20	20	30	294.96
30	30	20	20	305.32
30	30	30	10	299.34
30	40	20	10	315.21
40	20	20	20	339.88
40	20	30	10	329.23
40	30	20	10	342.07
50	20	20	10	394.85

We obtained that the maximum total efficiency occurs at the ratio $A_1 : A_2 : A_3 : A_4 = 50 : 20 : 20 : 10$.

Next, we run our model on an example generated arbitrary quadrilateral, consisting of these vertices: $A = (0, 0)$, $B = (32, 0)$, $C = (48, 50)$, and $D = (-8, 25)$, $\theta = 120^\circ$, and other parameters are the same. The selected results are as follows,

A_1	A_2	A_3	A_4	Maximum Total Efficiency
20	20	20	40	243.50
30	20	20	30	279.40
30	20	30	20	281.90
30	30	20	20	296.06
30	30	30	10	289.16
40	20	20	20	333.65
40	20	30	10	330.74
40	30	20	10	339.31
50	20	20	10	406.22

We also obtained that the maximum total efficiency occurs at the ratio $A_1 : A_2 : A_3 : A_4 = 50 : 20 : 20 : 10$. The figures below show the allocation of the land.



7 Discussions

Firstly, our results show that the magnitude of γ or the inclination is insignificant towards the height optimization algorithm, as the results are relatively the same across various steepnesses, simply fixing a certain γ value is sufficient to allocate by height.

Similarly, θ also has negligible impact on the total efficiency, as across various θ values, it mostly determines the direction in which the house will be located at, along with the rice field being at the opposite sides, hence, similar to rotating the zones around. Despite this, θ impacts the overall spread of the house, as the center of mass will be relatively high, but the house may spread on one side or compacted into a corner, both results in similar total efficiency.

Secondly, according to all the assumptions and constraints, the baseline results are not too far-fetched, as we aim to have sufficient water for year-round agricultural use, and only require the rice field and mixed crops to create self-sustainance for the farmers, not necessarily focused on highest yield or profit. Consequently, our model achieved this by having just enough residential area, rice field, and mixed crops

for sustainability, and the rest can be allocated towards having sufficient water resources, resulting in the 50 : 20 : 20 : 10 ratio, with many other ratios also being more effective than the given 30 : 30 : 30 : 10 ratio.

Subsequently, the results obtained from the other simulated quadrilaterals also align well within the baseline assessment, indicating that the $A_1 : A_2 : A_3 : A_4 = 50 : 20 : 20 : 10$ ratio is suitable and yields the highest total efficiency by a significant margin. Therefore, we propose the 50 : 20 : 20 : 10 ratio to be a flexible ratio in accordance to the core philosophy of the “New Theory”, prioritizing self-sustainance and suitability under the given considerations and assumptions. Our hard constraints such as the road connection have also been achieved

Nevertheless, due to the center of mass grouping optimizations, the water’s center of mass tends to be at the middle of the land plot, right between the rice field’s and mixed crops’ in order to minimize the distance between these center of masses, and results in the water resource may not be connected to the boundaries where the source of water inflow in at, and each zone may not be fully connected as well.

Thus, after the simulation by the model, we may conduct simple allocation modification processes, according to real-world scenarios, farmer’s own preferences, budget constraints, etc. later on, i.e. create small canals to connect the water resource with the sources, connect disconnected areas, adjust some areas for convenience constructions in the field, or shape the house based on architectural preferences. Which these modifications can be done based on the model’s final center of masses that serve as a weighted center of each area to maximize efficiency, and can be used easily to make fine adjustments based on the farmer’s decisions, while preserving the 50 : 20 : 20 : 10 as the center of our flexible ratio.

8 Model Assessment

8.1 Suitability

In essence, our model has assessed and considered various factors contributing to the land allocation suitability, which includes, but not limited to, the convenience for the farmers, efficient agricultural location for higher yields in suitable conditions, and overall that the relative positions reflect reasonable land allocations.

To begin, the model considered the shape of the water resource to be a frustum which reflect a commonly used ratio for pond shaping, meaning the model prioritizing in merging the water into a single

connected component, reasonable within our considerations. It also allocates the rice field and mixed crops to be close to the water for easy access and lowest cost in pipeline system implementation, ensuring all plants are hydrated properly, alongside the increased natural humidity seeping into surrounding soil.

Subsequently, we have also emphasized on the relative elevations between the plantations and the water, as the rice field requires a flooded field, while the mixed crops need less water, hence, the rice field is placed lower than the mixed crops, effectively reducing frequent water replenishments.

Furthermore, since it is required that cars should be able to access the residential area through the road, if the house were to be disconnected from the boundaries, building a road to connect it would be inefficient and reduce the overall efficiency, thus, the residential area will always have direct access to the road, and is at a high elevation as protection from floods.

8.2 Sustainability

Our model have also balanced between the suitability goals and sustainability goals, in accordance with the gist of the "New Theory", the stable and self-sustenance agricultural settings.

First and foremost, water sustainability is at the center of our attention. Farmers should have sufficient water that lasts a certain period in dry seasons or climate changes. Our model has not taken fine-scale weather factors into consideration. Due to high instability of rain cycles in a day, or a week, we instead constrained the pond volume to store the water enough for 60-day use without any additional input, which covers at least half the rice-culturing season, with the optimal goal of year-round sustainance.

Nonetheless, the rice field must also cover daily food consumptions of the farmers. We constrained the rice yield to exceed the average amount of rice consumed yearly by Thais, as reflected in water usage constants. And despite the sloped terrain, we have also modeled a simple method in determining sunlight blockage from the sun declination and the position of the residential area (supposed highest elevation) to ensure enough sunlight throughout.

Ultimately, our model has succeeded in assessing both sustainable and suitability goals up to some extent, which we believe is sufficient and significant for preliminary plannings in real-world land allocations, and would be of great usage while maintaining an easily understandable model for farmers.

8.3 Strengths

Our model has considered most of the physical needs for farmers, including plantation yields, water managements, and basic residential convenience. Thus, our model is versatile enough to be applied in real-world scenarios, while it remained simple enough to understand and interpret on real-world decision making.

Our model have also evaluated conditions within all the given constraints and assumption, including the inclination of the land in the west-east direction, implementation of geometry for dimension and area calculations, considered water availability, sunlight, and roadway access, and most importantly, simulate land allocations across categories of land usages with visualization of the final land diagram and a flexibility in the input parameters.

Significantly, our model can also be applied with a land plot of any N-sided convex polygon, since our code takes the polygon vertices and others as its input parameters, results have verified that larger, more complex, convex polygons are within the prediction scope of our model as well. Thus, our model is significantly more applicable in real-world scenarios where the land is not a perfect quadrilateral.

8.4 Weaknesses

Despite the strengths, our model has still got some weaknesses due to the additional considerations and assumptions given by the authors. One such assumption is that the model still assume many values to be constant, such as the steepness, the rate of water flowing in, or the rate of water usage per area. In real life, these values are not constant, and are mostly time-dependent both from the time of day and time of the year.

As the land initialization begins with complete randomness, with the sole constraint of fixed area ratio, the optimization sometimes stucks and the efficiency outcome is not proper enough. It rarely occurs, but it can interfere with ratio optimization enough to make the program not completely automatic.

9 Conclusion

The task provided to us is to extend the "New Theory" towards irregular quadrilateral of area 1-rai by using mathematical models. We have created a model composed of sub-goals that could determine the total efficiency of a particular land allocation, and developed a way to compute and visualize the most optimal allocation given just only the land corner ver-

tices, and the model is adaptable to any convex N-sided polygon as well.

We have also tested the model with example test-cases. The example results showed that the ratio of

$$A_1 : A_2 : A_3 : A_4 = 50 : 20 : 20 : 10$$

yields the highest total efficiency, and can be easily understood and adapted by farmers in designing their own land, based on their own preferences and additional need, for a sustainable and suitable land allocation by the New Theory Principles.

Appendix A: Analytical Hierarchy Process

The intensity of importance for the pair-wise comparisons are shown below,

Intensity of Importance	Classification
1	Equal
2	Weak or Slight
3	Moderate
4	Moderate-plus
5	Strong
6	Strong-plus
7	Very Strong
8	Very, Very Strong
9	Extreme

These importance ratings have been discussed and weighted given the sustainability and suitability goals, computed using the geometric mean of the authors opinions.

	$E_{d,2}$	$E_{h,2}$	$E_{l,2}$	AHP Weights
$E_{d,2}$	1	1/2	2	0.311905
$E_{h,2}$	2	1	2	0.490476
$E_{l,2}$	1/2	1/2	1	0.197619

	$E_{d,3}$	$E_{h,3}$	$E_{l,3}$	AHP Weights
$E_{d,3}$	1	2	3	0.548485
$E_{h,3}$	1/2	1	1	0.240909
$E_{l,3}$	1/3	1	1	0.210606

	W_1	W_2	W_3	W_4	AHP Weights
W_1	1	1	2	3	0.341649
W_2	1	1	3	3	0.377363
W_3	1/2	1/3	1	1	0.140494
W_4	1/3	1/3	1	1	0.125788

Appendix B: C++ Code Used for the Simulation of Land Allocation

```

/* New Theory Principles Land Allocation
   Modeling, an interactive simulation
*
* I, the sole author of this software,
  hereby declare that
* 1.) No part of this software, are
  generated nor derived from large
  language models,
* and AI-assisted autocomplete
  technologies.
* 2.) This software are to be released
  with MMC051 Math modeling submission.
*
* The MIT License
*
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*

```

```

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  CONTRACT, TORT OR OTHERWISE, ARISING
  FROM,
* OUT OF OR IN CONNECTION WITH THE SOFTWARE
  OR THE USE OR OTHER DEALINGS IN
  THE SOFTWARE.
*/

```

```

#pragma GCC optimize("Ofast")
#pragma GCC optimize("unroll-loops")
#pragma GCC target("avx2")
#include <bits/stdc++.h>
#include <cassert>
#include "Random.h"
#include "ANSI.h"
using namespace std;

#define Point complex<int>
#define PointD complex<float>
#define X real()
#define Y imag()
int Rwater, Rrice, Rcrop;

class Polygon {
public:
    int W, H;
    int verticesCount;
    vector<Point> vertices;

    void definePolygon(vector<Point>
vertices){
        verticesCount = vertices.size();
        this->vertices = vertices;
    }

    int orientationTest(Point P, Point P1,
Point P2)
    {
        return (P1.X-P.X)*(P2.Y-P.Y)-(P2.X-P.X
)*(P1.Y-P.Y);
    }
}

```

```

int isInsidePolygon(Point P)
{
    int count = 0;
    for (int i = 0; i < verticesCount; ++i)
    {
        int nextI = (i + 1) % verticesCount;
        if( (P.Y - vertices[i].Y) * (P.X - vertices[nextI].X) == (P.Y - vertices[nextI].Y) * (P.X - vertices[i].X))
        {
            if( min(vertices[i].X, vertices[nextI].X) <= P.X && P.X <= max(vertices[i].X, vertices[nextI].X)
            && min(vertices[i].Y, vertices[nextI].Y) <= P.Y && P.Y <= max(vertices[i].Y, vertices[nextI].Y) )
            {
                // Point is located on the boundary
                return 1;
                break;
            }
        }
        if (i == verticesCount - 1)
        {
            int count = 0;
            for (int j = 0; j < verticesCount; ++j)
            {
                int nextJ = (j + 1) % verticesCount;
                if(vertices[j].Y <= P.Y && P.Y < vertices[nextJ].Y && orientationTest(P, vertices[j], vertices[nextJ])>0) ++count;
                if(vertices[nextJ].Y <= P.Y && P.Y < vertices[j].Y && orientationTest(P, vertices[nextJ], vertices[j])>0) ++count;
            }
            return count % 2;
        }
    }
    return 0;
};

class Utilities
{
public:
    inline float toDegrees(float x){
        return x*numbers::pi/180;
    }

    inline float dist(int ch1,int ch2,vector<PointD> vec){
        return sqrt((vec[ch1].X-vec[ch2].X)*(vec[ch1].X-vec[ch2].X)+(vec[ch1].Y-vec[ch2].Y)*(vec[ch1].Y-vec[ch2].Y));
    }
};

class ZonePolygon: public Polygon, public Utilities {
public:
    int grid[107][107];
    int zone[107][107];

```

```

int Z, theta, gamma;
float sinTheta, cosTheta, tanGamma;
float dmax=0, hmax=0;
vector<Point> latticePoints;

vector<PointD> centerOfMass;
vector<float> centerOfHeight;
vector<float> Ai;

inline float h(PointD P){
    return P.X * sinTheta * tanGamma + P.Y * cosTheta * tanGamma;
}

void zoneDivision()
{
    int count = 0;
    for (int x = 0; x < W; ++x)
    {
        for (int y = 0; y < H; ++y)
        {
            int location = isInsidePolygon(Point(x, y));
            int rng=Random::get(1,100);
            if(location)
            {
                grid[x][y] = location;
                latticePoints.push_back(Point(x, y));
                hmax = max(hmax, h(PointD(x,y)));
            };

            if(rng<=Rwater) rng=1;
            else if(rng<=Rwater+Rrice) rng=2;
            else if(rng<=Rwater+Rrice+Rcrop) rng=3;
            else rng=4;

            zone[x][y]=rng;
            Ai[rng]++;
            count++;
        }
    }

    for(int i=0;i<latticePoints.size();++i)
    {
        /*
        int rng;
        if(i<0.4*latticePoints.size()) rng = 2;
        else if(i<0.7*latticePoints.size()) rng = 1;
        else if(i<0.9*latticePoints.size()) rng = 3;
        else rng = 4;
        zone[latticePoints[i].X][latticePoints[i].Y]=rng;
        */
    }

    for(auto i:vertices){
        for(auto j:vertices){
            dmax=max(dmax,(float)sqrt((i.X-j.X)*(i.X-j.X)+(i.Y-j.Y)*(i.Y-j.Y)));

```

```

    }
}
}

int depth[107][107];
float V;
inline void updateCenter(){
    fill(Ai.begin(),Ai.end(),0);
    queue<pair<pair<int,int>,int>> q;
    vector<float> sx,sy,sh;
    sx.resize(Z+1); sy.resize(Z+1); sh.
resize(Z+1);
    for(int i=0;i<W;++i){
        for(int j=0;j<H;++j){
            depth[i][j]=0;
            if (zone[i][j] == 1)
            {
                if(i == 0 || i== W-1 || j == 0
|| j == H-1){
                    depth[i][j] = 1;
                }
                else {
                    for (int r1 = -1; r1 <= 1; ++
r1){
                        for (int r2 = -1; r2 <=1 ;
++r2){
                            if(zone[i+r1][j+r2]!=zone[
i][j])
                                {
                                    depth[i][j]=1;
                                }
                            }
                        }
                    }
                    if(depth[i][j])
                    {
                        q.push({{i,j},1});
                    }
                }
            }
            sx[zone[i][j]]+=i;
            sy[zone[i][j]]+=j;
            sh[zone[i][j]]+=h(PointD(i,j));
            ++Ai[zone[i][j]];
        }
    }
    for(int i=1;i<=Z;++i){
        sx[i]/=Ai[i]; sy[i]/=Ai[i]; sh[i]/=
Ai[i];
        centerOfMass[i]={sx[i],sy[i]};
        centerOfHeight[i]=sh[i];
    }
    while(!q.empty()){
        int t1 = q.front().first.first;
        int t2 = q.front().first.second;
        int t3 = q.front().second;
        q.pop();
        for (int r1 = -1; r1 <= 1; ++r1){
            for(int r2 = -1; r2 <= 1; ++r2){
                if(t1+r1>=W || t2+r2>=H || t1+r1
<0 || t2+r2<0) continue;
                if(zone[t1+r1][t2+r2]!=1)
continue;
                if(!depth[t1 + r1][t2 + r2]){
                    depth[t1 + r1][t2 + r2] = t3 +
1;
                    q.push({{t1 + r1, t2 + r2}, t3
+ 1});
                }
            }
        }
    }
}

```

```

    }
}
}
V = 0;
for(int i=0;i<W;++i){
    for(int j=0;j<H;++j){
        V += depth[i][j]/3.0;
    }
}
};
class Land: public ZonePolygon
{
public:
    int DR, DW, Rin;
    float U2 = 0.0090, U3 = 0.0045;
    float dayOpt = 365, kW1 = 7.4511e-6;
    float Wd2 = 0.3119, Wh2 = 0.4905, W12 =
0.1976;
    float Wd3 = 0.5485, Wh3 = 0.2409, W13 =
0.2106, kh3wohmax = -log(0.5);
    // Initialization
    Land(vector<Point> vertices, int theta,
int gamma, int DR, int DW, int Rin){
        definePolygon(vertices);
        this->W = 50;
        this->H = 80;
        this->Z = 4;
        this->theta = theta;
        this->gamma = gamma;
        this->DR = DR;
        this->DW = DW;
        this->Rin = Rin;
        sinTheta = sin(toDegrees(theta));
        cosTheta = cos(toDegrees(theta));
        tanGamma = tan(toDegrees(gamma));
        centerOfMass.resize(Z+1);
        centerOfHeight.resize(Z+1);
        Ai.resize(Z+1);
        zoneDivision();
        updateCenter();
    }

    // Print

    inline void print(){
        ANSI::reset();
        for(int i=0;i<W;++i){
            for(int j=0;j<H;++j){
                if(zone[i][j]==1) ANSI::color(34);
                if(zone[i][j]==2) ANSI::color(33);
                if(zone[i][j]==3) ANSI::color(32);
                if(zone[i][j]==4) ANSI::color(31);
                //ANSI::color(30+zone[i][j]);
                if(zone[i][j]==0) cout << ' ';
                else cout << zone[i][j];
            }
            cout << endl;
        }
        ANSI::color(0);
        updateCenter();
        cout << "STATISTIC " <<
totalEfficiency(centerOfMass,
centerOfHeight, V, Ai) << " " << endl;
        for(auto i:Ai) cout << i << " ";
        cout << "W VALIDITY " << (V>=60*U2*Ai
[2]+60*U3*Ai[3]);
        /*for(int i=0;i<W;++i){

```

```

        for(int j=0;j<H;++j) cout << depth[i][j];
        cout << endl;
    }*/
}

// Optimizer
float prev = 0;
int counter = 0;
inline void optimizer(int time){
    updateCenter();
    auto oldc1 = centerOfMass;
    auto oldc2 = centerOfHeight;
    auto oldc3 = V;
    auto oldc4 = Ai;

    vector<int> a(time),b(time),ax(time),ay(time),bx(time),by(time),var(time);
    for(int i=0;i<time;++i){
        a[i] = Random::get(0,latticePoints.size()-1);
        b[i] = Random::get(0,latticePoints.size()-1);
        ax[i]=latticePoints[a[i]].X; ay[i]=latticePoints[a[i]].Y;
        bx[i]=latticePoints[b[i]].X; by[i]=latticePoints[b[i]].Y;

        if(zone[ax[i]][ay[i]] !=3 ){
            do
            {
                int dx = (ax[i]-centerOfMass[zone[ax[i]][ay[i]]].X>0)? -1:1;
                int dy = (ay[i]-centerOfMass[zone[ax[i]][ay[i]]].Y>0)? -1:1;
                dx *= Random::get(0,20);
                dy *= Random::get(0,20);
                bx[i]=ax[i]+dx,by[i]=ay[i]+dy;
            } while(bx[i]<0 || by[i]<0 || bx[i]>=W || by[i]>=H || find(latticePoints.begin(),latticePoints.end(),Point(bx[i],by[i]))==latticePoints.end());
            }
            var[i]=Random::get(1,Z);
            swap(zone[ax[i]][ay[i]],zone[bx[i]][by[i]]);
        }

        ///swap(zone[ax][ay],var);
        updateCenter();
        auto newc1 = centerOfMass;
        auto newc2 = centerOfHeight;
        auto newc3 = V;
        auto newc4 = Ai;
        // Constraints
        //

        if(totalEfficiency(newc1, newc2, newc3, newc4)>=totalEfficiency(oldc1, oldc2, oldc3, oldc4)){
            // Favorable
            prev = V;
            counter = 0;
        }
        else{
            for(int i=time-1;i>=0;--i){
                swap(zone[ax[i]][ay[i]],zone[bx[i]

```

```

]][by[i]]);
            }

            /*
            //swap(zone[ax][ay],zone[bx][by]);
            if(counter>10){
                swap(zone[ax][ay],zone[bx][by]);
                //swap(zone[ax][ay],var);
            }
            else{
                prev=V;
            }
            ++counter;
            */
        }
    }

    // Model

    inline float totalEfficiency(vector<PointD> centerOfMass, vector<float> centerOfHeight, float V, vector<float> Ai){
        if(!(Rin>=U2*Ai[2]+U3*Ai[3])) return -1;
        // 3.5.1 Water mechanics
        //if(!(V>=60*U2*Ai[2]+60*U3*Ai[3])) && V<prev) return -1;
        // 3.5.2 Volume
        float Atotal = Ai[1]+Ai[2]+Ai[3]+Ai[4];
        if(!(Ai[2]/Atotal>=300.00/1600.00)) return -1;
        if(!(Ai[3]>=200.00/1600.00)) return -1;
        if(!(Ai[4]>=100.00/1600.00)) return -1;
        float statistic = 0;
        for (int zone = 1; zone <= Z; ++zone)
        {
            float Wi, Ei;

            if(zone == 1)
            {
                float dayActual = V / (U2 * Ai[2] + U3 * Ai[3]);
                Ei = exp(-kW1*(dayActual - dayOpt))*(dayActual - dayOpt));
                Wi = 0.3416;
            }
            else if(zone == 2){
                float d21=dist(2,1,centerOfMass);
                float Ed2 = 1 - abs(d21/dmax);
                float Eh2 = 1 - centerOfHeight[2]/hmax;
                float phi2 = 90 + atan(toDegrees((centerOfMass[4].Y-centerOfMass[2].Y)/(centerOfMass[4].X-centerOfMass[2].X)));
                float El2 = phi2/180;
                Ei = (Ed2*Wd2+Eh2*Wh2+El2*Wl2)/(Wd2+Wh2+Wl2);
                Wi = 0.3774;
            }
            else if(zone == 3){
                float d31=dist(3,1,centerOfMass);
                float Ed3 = 1 - abs(d31/dmax);
                float Eh3 = exp(-kh3wohmax/(hmax*hmax/4.0)*(centerOfHeight[3]-(hmax/2))*(centerOfHeight[3]-(hmax/2)));

```

```

        float phi3 = 90 + atan(toDegrees((
centerOfMass[4].Y-centerOfMass[3].Y)/(
centerOfMass[4].X-centerOfMass[3].X)));
        float El3 = phi3/180;
        Ei = (Ed3*Wd3+Eh3*Wh3+El3*Wl3)/(
Wd3+Wh3+Wl3);
        //cout << d31 << " " << Ed3 << " "
        << Eh3 << " " << phi3 << " " << El3 <<
        " " << Ei << endl;
        Wi = 0.1405;
    }
    else if(zone == 4){
        Ei = centerOfHeight[4]/hmax;
        Wi = 0.1258;
    }
    else{
        Ei = 0;
    }
    //cout << Ei << " ";
    statistic += Ai[zone] * Wi * Ei;
}
//cout << statistic << " " << endl;
return statistic;
}

};

signed main(){
    cout << "Input theta, gamma, iterations: "
    ;
    int theta, gamma, iterations, temp1, temp2
    ; cin >> theta >> gamma >> iterations;
    cout << "Input Rwater, Rrice, Rcrop,
    Rresidential (sum = 100)";
    cin >> Rwater >> Rrice >> Rcrop >> temp1;
    int N; cin >> N;
    vector<Point> input;
    for (int i = 0; i < N; ++i){
        cin >> temp1 >> temp2;
        input.push_back(Point(temp1, temp2));
    }
    Land farm(input,theta,gamma,0,0,300);
    farm.print();
    for(int i=0;i<iterations;++i){
        farm.optimizer(2);
        if(i%1000==0) farm.print();
    }
    farm.print();
}

```

Listing 1: code.cpp

```

        escape();
        std::cout << "H" << std::flush;
    }
    void color(int id){
        escape();
        std::cout << std::to_string(id) << "m"
        ;
    }
    void code(std::string str){
        escape();
        std::cout << str;
    }
}

```

Listing 2: ANSI.h

```

/*
 * Random number generator based on Mersenne
 * Twister
 * Random.h
 * Released under the MIT License, MMC051
 * Contributor
 */
#pragma once
#include <random>

namespace Random {
    inline std::mt19937& engine() {
        static thread_local std::mt19937 eng
        ([[]() {
            std::random_device rd;
            return std::mt19937(rd());
        }())
        ;
        return eng;
    }

    inline int get(int min, int max) {
        std::uniform_int_distribution<int>
        dist(min, max);
        return dist(engine());
    }
}

```

Listing 3: Random.h

```

/*
 * ANSI escape sequences for Linux terminal
 * emulators
 * ANSI.h
 * Released under the MIT License, MMC051
 * Contributor
 */
#pragma once
#include <iostream>
#include <string>
namespace ANSI {
    void escape(){
        std::cout << "\033[";
    }
    void reset() {

```

Appendix C: Simulation Results and Assessments

The generic square baseline model:

A_1	A_2	A_3	A_4	Maximum Total Efficiency
20	20	20	40	260.213
20	20	30	30	258.612
20	20	40	20	Insufficient Water
20	20	50	10	Insufficient Water
20	20	60	0	Insufficient Water
20	30	20	30	Insufficient Water
20	30	30	20	Insufficient Water
20	30	40	10	Insufficient Water
20	30	50	0	Insufficient Water
20	40	20	20	Insufficient Water
20	40	30	10	Insufficient Water
20	40	40	0	Insufficient Water
20	50	20	10	Insufficient Water
20	50	30	0	Insufficient Water
30	20	20	30	294.96
30	20	30	20	289.87
30	20	40	10	282.12
30	20	50	0	Insufficient House
30	30	20	20	305.32
30	30	30	10	299.34
30	30	40	0	Insufficient House
30	40	20	10	315.21
30	40	30	0	Insufficient House
30	50	20	0	Insufficient House
40	20	20	20	339.88
40	20	30	10	329.23
40	20	40	0	Insufficient House
40	30	20	10	342.07
40	30	30	0	Insufficient House
40	40	20	0	Insufficient House
50	20	20	10	394.85
50	20	30	0	Insufficient House
50	30	20	0	Insufficient House
60	20	20	0	Insufficient House

An arbitrary quadrilateral model, with the vertices as (0,8), (0,40), (50,56), (25,0), yields the following result:

A_1	A_2	A_3	A_4	Maximum Total Efficiency
20	20	20	40	243.50
20	20	30	30	Insufficient Water
20	20	40	20	Insufficient Water
20	20	50	10	Insufficient Water
20	20	60	0	Insufficient House
20	30	20	30	Insufficient Water
20	30	30	20	Insufficient Water
20	30	40	10	Insufficient Water
20	30	50	0	Insufficient House
20	40	20	20	Insufficient Water
20	40	30	10	Insufficient Water
20	40	40	0	Insufficient House
20	50	20	10	Insufficient Water
20	50	30	0	Insufficient House
30	20	20	30	279.40
30	20	30	20	281.90
30	20	40	10	272.223
30	20	50	0	Insufficient House
30	30	20	20	296.06
30	30	30	10	289.16
30	30	40	0	Insufficient House
30	40	20	10	Insufficient Water
30	40	30	0	Insufficient House
30	50	20	0	Insufficient House
40	20	20	20	333.65
40	20	30	10	330.74
40	20	40	0	Insufficient House
40	30	20	10	339.31
40	30	30	0	Insufficient House
40	40	20	0	Insufficient House
50	20	20	10	406.22
50	20	30	0	Insufficient House
50	30	20	0	Insufficient House
60	20	20	0	Insufficient House

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